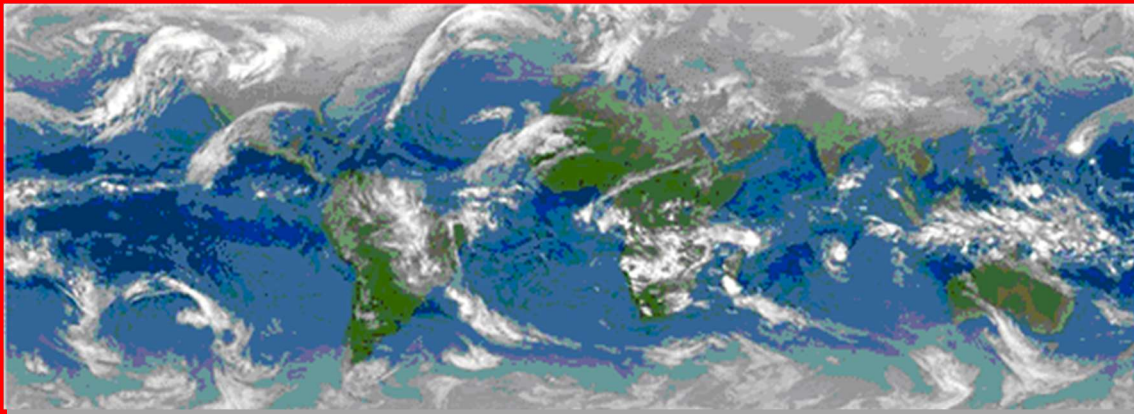


ESA EarthCODE



Platform Engineering and Product Update Services Provision

Ref: EARTH-CODE.BPD.08
Version 1.0

Date: 06/07/2026

For: ESA ESRIN
Ref: AO/I-11828/23/I-NS TBD



SIGNATURES AND COPYRIGHT

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DOCUMENT AMENDMENT RECORD

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0	1	10/03/2026	Garin Smith		V0.1 service beta delivery
1	0	06/07/2026	Garin Smith		V1 initial service live delivery

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1 INTRODUCTION

1.1 Executive Summary

This document defines the formal process to provide engineering support services and science product update services to EarthCODE customers.

Engineering support services concern the provision of engineering support to help publish Data (Products), Experiments and Workflows. They can also be used to make modifications to workflows, so that they can be accepted for use by the Science Products Update Service to create new products.

Science product update services concern the provision of new products by a platform using an existing workflow. They support the provision of an **operational service** against a defined SLA. This service can configure an existing workflow to provide new products, but workflow software changes should be supplied by the Engineering Support Definition Service.

In later sections we formally define the following

- D.1. Engineering Support Service Definition (status operational)
- D.2. Engineering Support Service Report (status operational)
- D.3. Science Products Update Service Definition (status operational)
- D.4. Service Operations Report (status operational)

These sections outline the formal process to request and access these services and the process to report on implementation progress. They define how these services will be provided and will be updated as required as service provision changes.

Service requests are made and monitored using GitHub projects and this is also described in the sections below.

- [Engineering Support · Engineering and Product Update Services](#) (status operational)

1.2 Applicable Documents

The following documents are contractually applicable to the content of this document and are denoted AD-n, where 'n' is the number in the list below:

AD	Title
AD 1.	Statement of Work (SoW), ESA-EOP-SD-SOW-0418, Issue 1, Revision 0, 08/06/2023, ref EARTH-CODE.BPD.02
AD 2.	CHANGE REQUEST No. 3 to Contract No. ESRIN Contract No. 4000143045/23/I-NS, ref: ESA-EOP-SG-CCR-0743

1.3 Reference Documents

The following documents are referenced in this document and are denoted RF-n, where 'n' is the number in the list below:

Ref	Title/Source
RF 1.	EarthCODE State of the Art Analysis and Architecture Development Roadmap

Ref	Title/Source
RF 2.	EarthCODE Security Management Plan
RF 3.	EarthCODE Operations Manual
RF 4.	Contributing to the Open Science Catalogue — earthcode
RF 5.	Build STAC Metadata For A COG — earthcode
RF 6.	Zarr Metadata With xstac — earthcode
RF 7.	GeoParquet STAC Metadata — earthcode
RF 8.	Data Best Practices EarthCODE

1.4 Acronyms and Abbreviations

Please refer to EarthCODE State of the Art Analysis and Architecture Development Roadmap (RF 1.) for a list of Acronyms and Abbreviations.

1.5 Definition of Terms

Please refer to EarthCODE State of the Art Analysis and Architecture Development Roadmap (RF 1.) for a list of Terms.

2 D.1. ENGINEERING SUPPORT SERVICE DEFINITION

This service concerns the provision of engineering support to help publish Data (Products), Experiments and Workflows. It also includes support to help improve the FAIR score of Data (Products), Experiments and Workflows prior to publication. In addition, it can also be used to make modifications to workflows so that they can be accepted for use by the Science Products Update Service to create new products.

Requests for this service are likely to come from ESA or the EarthCODE consortium using the following link [Engineering Support · Engineering and Product Update Services](#). Requests may also originate from personal interactions at Conferences, or other personal interactions including emails.

2.1 Engineering Support Service Lifecycle

The following Engineering Support Service Lifecycle is used:

1. Engineering Support request is added to the board ([Engineering Support · Engineering and Product Update Services](#))
2. Request is authorized by ESA
 - 2.1. ESA sends WO request by email (may include from the start the details provided in the form and agreed with the subcontractors involved)
 - 2.2. Offer is provided to ESA by TPZ via response to WO, including the cost.
 - 2.3. The WO is signed by ESA TO, CO and TPZ, and exchanged by email.
3. Request is fulfilled and the GitHub project is used to provide progress updates by the service provider
4. Request is completed and a Product, Workflow, or Experiment is published or pre-published to the live EarthCODE environment

The work order procedure is defined in the appendix to CCN-4, an excerpt of this is provided for information in Appendix A – Work Order Procedure

The Service Manager role is responsible for managing this overall lifecycle.

We will support the following status options used in the diagram below:

Status	Description
Request	An initial service request for help from or on behalf of a scientist that is ready for review.
Review Request	ESA have authorised the request and the Service Manager can then review the task size and further details as need. Typically, requests that are in budget and support publication of ESA approved projects are considered favourably.
Reject	Following ESA review, it is decided no further action required – Rejection can be on the basis of budget or scope not meeting ESA requirements.
Todo	After Service Manager review, a Subcontractor can then decide if they want to do the work request.
Accept Request	A Subcontractor can accept a request to do the work, and the Service Manager can then Populate a WO.
Change Request	A Subcontractor can suggest changes to the work requested and the review process above can be repeated.
WO To Be Signed	ESA Can create formal WO so that all parties can sign it.
In Progress	The WO is signed by ESA TO, CO and TPZ, and exchanged by email so that the Subcontractor can start work.
Done	A service request is considered complete by a Subcontractor and ready for review by the Service Manager.

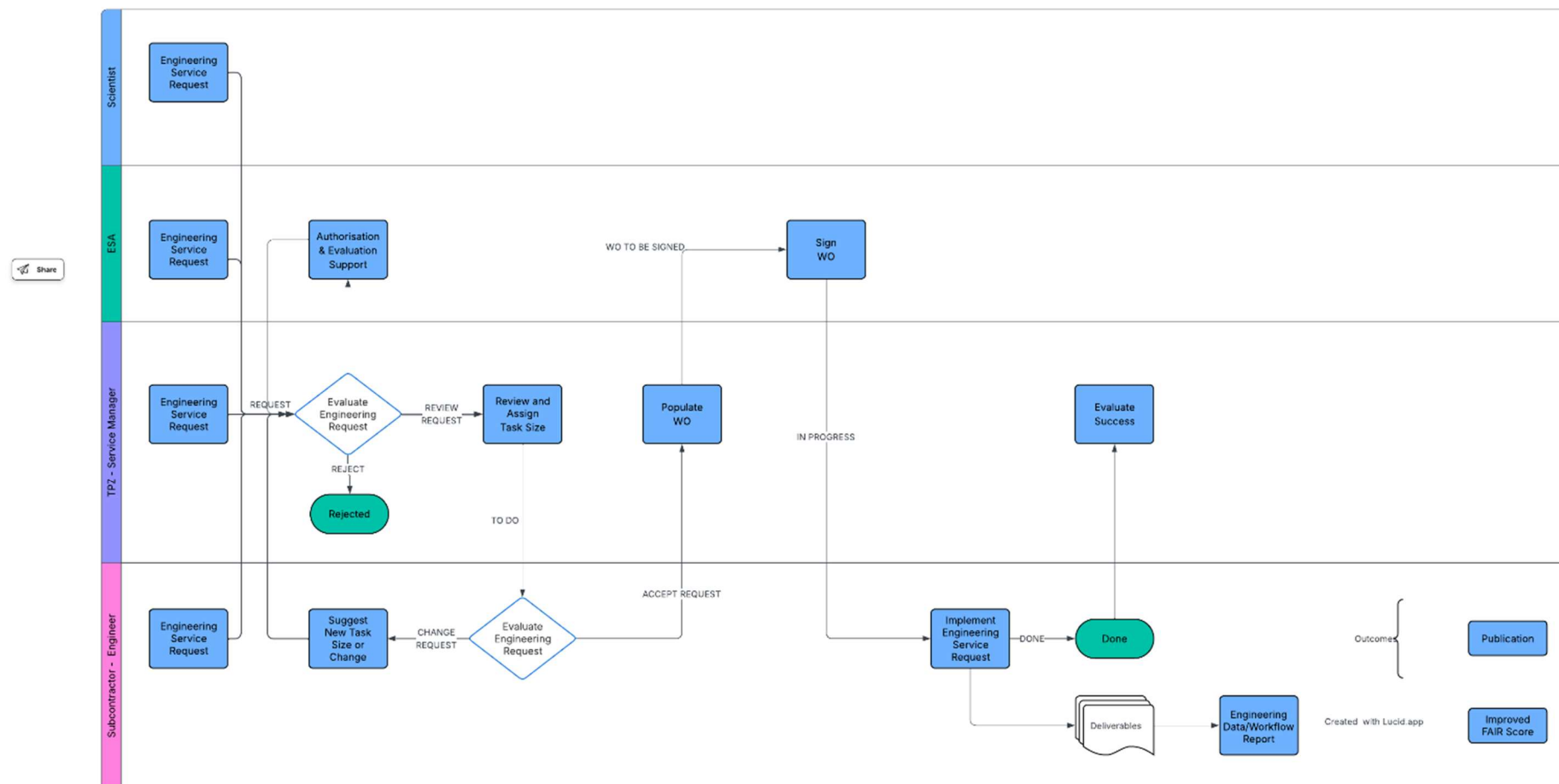


Figure 1 : EarthCODE - Engineering Support Service Lifecycle

2.2 EarthCODE Acceptable Formats

These formats are acceptable for publishing to EarthCODE. They are also acceptable to the EarthCODE PRR (Projects Results Repository) solution.

Please note that RF 1 (EarthCODE State of the Art Analysis and Architecture Development Roadmap) includes an ICD (Interface Control Document) that defines how valid experiments are defined. This includes schema definitions. Please refer to Validate Experiment and Validate Product.

EarthCODE Acceptable Formats Summary	
Data type	Preferred format
Raster scenes & mosaics	COG (GeoTIFF) with internal tiling and overviews for fast partial reads or Zarr; lossless compression by default.
n-D data cubes (time/lat/lon/level)	Zarr v3/GeoZarr with sensible chunking and consolidated metadata
NetCDF Virtualized	kerchunk (JSON references) or VirtualiZarr to present existing netCDF-4/HDF5 or GRIB2 as a virtual Zarr for efficient, chunked cloud reads without rewriting source data.
Vector Data	FlatGeobuf with spatial index; suitable for HTTP streaming and subsetting ; alternatively GeoParquet.
Tabular in-situ/model outputs	Parquet (preferred) or CSV for small datasets; define schema, units, and time zones.
Other type	Other formats can also be used when approved by ESA on a case by case basis.
Workflow type	Preferred Format
Application Package	CWL in Package Registry, Application Package in container registry
OpenEO	An openEO process graph (experiments) or user defined process (UDP) definition (workflow) in JSON format, shared in a public repository.
Jupyter Notebook	A Jupyter Notebook together with the kernel to execute it
Argo workflow	Argo workflow description
Other type	Other formats can also be used when approved by ESA on a case by case basis.

2.3 Engineering Support Services

Workflow maturity and complexity	Service Description (D1-D8 and W1-W7 definition provided later)	Size	Range for Days of Work (can be converted to hours)
	Metadata Creation and Publishing – for workflow and data, including asset level metadata - (D1-D3) & (W1-W2). Optionally, if the data is in the correct format and can be visualised this includes Data Upload (D4).	S	1 to 3 days
	In addition to supporting them with publishing (all tasks under S) - work that involves light touch workflows/data engineering – D4-D6 and/or W2-W4. Light touch would imply that the scientists are hands-on working with you to support the transformation.	M	2 to 5 days
	In addition to M, when the code or data changes are more complex and the user requires more support.	L	5 to 20 days
	In addition to the tasks under L, packaging the product and workflow into a reproducible experiment and modifying the code so that it is ready for operational use within the defined scope. This means only configuration changes are then needed to run a workflow operationally and generate new products and no further code changes are required. This includes tasks D1-D8 and W1-W6. This would include delivering documentation on configuration, testing, proof of robustness (e.g. test coverage report) and confirmation of readiness for operational usage. In exceptional circumstances, this could also apply to very complex data engineering work.	XL	20 to 90 days

2.4 Data FAIRification & Publishing Services Definition

Data	
Data FAIRification & Publishing	
Task	Description
D1- Onboarding, Assessment And Communication	Understand the dataset, primary user group; agree the publishing plan (formats, metadata). Communicate and coach on cloud-optimised formats and FAIR/Open Science principles so teams can sustain changes in their own pipelines and organisation. Create, update, and maintain work in tickets and track communications. Note that this needs to be proportional to the amount of changes required for the data to be published. For CDSE, this task involves talking to the project team about the dataset that they aim to generate. This might impact the openEO implementation approach to optimize the result. The generation of

	cloud-optimized formats and complying to FAIR and Open Science principles is done automatically by the platform. DeepESDL: Consult data owner about available and recommended options, particularly a tailoring towards the main applications and user groups
D2- STAC-Asset-Level Metadata Creation	Create STAC Item + asset-level metadata so the dataset is discoverable and assets are usable (catalog-level STAC for discovery; file/asset details to interpret what each asset is and how to use it), following best practices and PRR requirements (https://esa-earthcode.github.io/tutorials/index-1/); following best practices - https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/FAIR%20and%20Open%20Science%20Best%20Practices/ - For CDSE, this task is done automatically by using openEO in combination with the EarthCODE tooling for publishing products, workflows, and experiments. - For DeepESDL, this task is assisted by deep-code.
D3- OSC STAC Collection-Level Metadata Creation & Publishing	Gather license/citation/description/provider and other details, advising users about possible options for licensing, DOI provisioning from EarthCODE and other required information (https://esa-earthcode.github.io/tutorials/index-2/); Create Open Science Catalogue STAC collections and publish them to the Open Science Catalogue GitHub Repository (https://github.com/ESA-EarthCODE/open-science-catalog-metadata) - following best practices https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/FAIR%20and%20Open%20Science%20Best%20Practices/ - Except for some of the advising discussions, this activity is part of the EarthCODE publication process through the EarthCODE tooling on CDSE and should not require additional efforts. - For DeepESDL, collection-level STAC metadata is created and published to the Open Science Catalogue via a deep-code pull request.
Data Engineering	
Task	Description
D4- Data Upload CloudFerro & PRR	Upload the product to the project storage and ensure it is archived/persisted in the ESA Projects Results Repository (PRR). We intend to automate PRR deployment, but it is possible that on occasions some additional support is needed. - Platform specific example one - For DeepESDL, this task is assisted by the automated publishing service provided by EarthCODE following the latest technical discussions with the consortium.
D5- Data Engineering & Optimization	Based on the assessment (and user agreement), transform data to EarthCODE-preferred cloud-ready formats and layouts (e.g., cloud-optimised rasters, chunked n-D cubes, see summary guide below), choosing chunking/tiling to match access patterns, optimal compression and other best practices for data outlined in EarthCODE: https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/Data%20and%20Workflow%20Best%20Practices/Data/.

	- This activity should already be covered through the usage of openEO, generating cloud optimized rasters. - Transformation is done leveraging existing tools from Python's data science stack, particularly xcube, zappend, and xrlint.
D6- Data Metadata Enrichment	Ensure embedded/file-level metadata is complete and consistent: CRS/spatial references, global attributes (e.g., title/summary/provenance/license), variable attributes (units/long_name), and CF-aligned variable naming/flags so datasets validate cleanly and are analysis-ready. (https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/Data%20and%20Workflow%20Best%20Practices/Data/) - This activity should already be covered through the EarthCODE tooling on CDSE that ensures that the published metadata is compliant with the EarthCODE schemas. - DeepESDL: Enforced by deep-code.
D7- Data Visualization / Dashboard Creation	Create visualisations or dashboards aligned to the user tasks identified during onboarding. - On CDSE the visualisation and dashboards are only supported through the existing tooling, such as the openEO Web Editor and the openEO integrations for Jupyter and QGIS. We do not foresee custom development of visualisations or dashboards. - DeepESDL: Optional step, can be supported via DeepESDL viewer or any other service using xcube Server as back-end (or client-side rendering). - Dedicated or shared eodash instances can be provided like for instance the Science Hub Dashboard or the Antarctica InSync Dashboard. - Lexcube.
D8- Tutorial Contribution	Contribute a working example showing how to access and use the data (and, where relevant, how to format/transform it for cloud use), on EarthCODE's tutorial page. (https://esa-earthcode.github.io/tutorials/) - This activity involves publishing a Jupyter Notebook that showcases a relevant use case of the data, generated through the CDSE openEO Federation. Preferably, this is done with support of the service requester to ensure the use case is relevant and correct. - For DeepESDL, tutorial contributions take the form of Jupyter Notebooks executed within the DeepESDL JupyterHub environment, demonstrating access to and analysis of published Zarr datacubes using xarray, xcube, and the broader Pangeo ecosystem.

2.5 Workflows and Experiments Publishing Services Definition

Workflows and Experiments	
Workflow FAIRification & Publishing	
Task	Description
W1- Onboarding, Assessment and Communication	Understand the workflow (note that this needs to be proportional to the amount of changes required for publishing), users, and potential target EarthCODE platform; agree if it can be reproducible and what that means for this workflow (inputs, parameters, environment, expected outputs), and produce a short publishing plan. Discussing the presented use case with the user and its implementation on in openEO on CDSE. This discussion should clearly define the scope of the activity, the expected outcomes, budget, planning, definition of roles and responsibilities and ways of collaboration.
W2- OSC FAIRification of Workflow & Publishing	Create OGC API records for the workflow following EarthCODE's Workflows metadata best practices https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/FAIR%20and%20Open%20Science%20Best%20Practices/Workflows; Fork repository so that it sits under an EarthCODE owned GitHub organisation. Publish OGC API Record to Open Science Catalog. - This is part of the automated flow supported through the EarthCODE publishing tooling on CDSE and should not require any additional effort. - For DeepESDL, OGC API records for the workflow are generated and published to the Open Science Catalog via a deep-code pull request
Workflow Engineering	
Task	Description
W3- Workflow packaging (openEO, OGC App Package, CWL, etc.)	Package or port the workflow so others can execute it reliably locally and/or on supported platforms (containerisation where appropriate, openEO processes, OGC App Packages, CWL, etc.), including pinned dependencies and clear run instructions. - This task involves taking a non-openEO-based algorithm (e.g. algorithm implementation in plain Python) and porting it to an openEO-based implementation. This activity requires a good understanding of the algorithm to translate its concepts to the usage of pre-defined functions and user-defined functions (UDF) on openEO. These exercises almost always require a custom approach and estimation effort if projects request additional engineering support. Examples: - Transferring an index calculation based on band math is easily transferable as openEO supports band math calculation using its predefined functions. This is well documented and several examples are available. In this case, engineering support might not be necessary for W3. - Inference of a pre-trained model might require retraining the model to a different format to be able to plug it into an openEO-based process, require additional validation of new model.

	○ If the algorithm is not aligned with datacube concepts, a porting exercise might require a very tailored approach. • For DeepESDL, the preferred workflow input is a Jupyter Notebook in a public Git repository with a clear README, defined execution environment (e.g. environment.yml or requirements.txt), and explicitly parameterised inputs. In addition, we could convert a Jupyter notebook into an EOAP on cases which requires this.
W4- Workflow onboarding & Testing Execution	Create/validate the runtime environment (Conda/mamba, container image, etc.) on the target platform and run end-to-end tests; ensure published metadata includes the right links/entry points to open/run the workflow on-platform • The runtime environment creation and validation is hidden by using openEO on CDSE, that eliminates the need to set up a runtime environment. Workflow metadata checks should be covered by the FAIR score tooling and the automated publishing process provided by the EarthCODE tooling on CDSE. Running e2e tests requires setting up execution scenarios on openEO that takes a workflow/experiment, parameters and a validated result as input and validates if the generated output matches with the validated result. • DeepESDL: Runtime environment is created automatically and the published can be executed for testing and demonstration.
W5- OSC Experiment Creation and Publishing	Gather and persist all experiment inputs (input datasets/products + parameter/config files + workflow reference) by uploading to CloudFerro/PRR, then publish an Experiment as an OGC API Records record to OSC GitHub, ensuring links for workflow + inputs + configuration + environment + outputs product (including distinct links for input/config and environment artifacts) - following EarthCODE best practices. Verify and test that experiment is executable. • For CDSE, the generation of the metadata and publication to the OSC GitHub is part of the publishing tooling provided by CDSE. No additional action should be required. • For DeepESDL, experiment inputs (input Zarr datasets, configuration files, workflow reference) are persisted on S3, and the experiment record is published to the OSC GitHub via a deep-code pull request.
W6 - Modify for Operational use	Modify a workflow, so that it can be used to create new Products by the Science Products Update Service Definition. The Products by the EarthCODE Science Products Update Service Definition can then be used to provide an operational solution using the workflow. The Science Products Update Service Definition will not be expected to make code changes, but it could make necessary config changes. Improve structure, readability, and reusability to follow recommended best practices for code and workflows on EarthCODE. (https://esa-earthcode.github.io/documentation/Community%20and%20Best%20Practices/Data%20and%20Workflow%20Best%20Practices/Workflows/) • For CDSE and the usage of openEO, this would be part of W1, W4 and/or W7. If this need is relevant from the start, this is taken into account for W4. If the workflow already exists and would not support systematic processing, this is considered part of W7. • For DeepESDL, this includes the re-engineering of the Jupyter Notebook into a proper Python package, suitable for scalable and robust execution.

W7 – Workflow optimization	Update an existing workflow to enhance its execution on a specific platform or for operational use as defined by the Products Update Service Definition. The objective is to adapt the current workflow, so it runs efficiently on the platform, reducing costs and boosting performance. This process also involves collecting benchmarks both before and after implementation. This exercise does not focus on improving the scientific performance of the algorithm. - For CDSE, this involves setting up benchmarks of the current services to track the current service metrics. Depending on the optimization scenario, discussed in W1 (e.g. lowering cost, lowering execution time, ...), a detailed analysis of the existing workflow is required. These exercises almost always require a custom approach and estimation effort. - For DeepESDL, this involves systematic profiling and benchmarking by experts as well as automated tools to improve code performance.
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2.6 Customer Engineering Support Request Form

The following template is used to request Engineering Support:

https://github.com/ESA-EarthCODE/OSC-Entries/blob/main/.github/ISSUE_TEMPLATE/engineering-support-request-template.md

It is based on the form below and can be updated further as the service evolves.

Engineering Request ID	ENG0001
Summary	
Request Summary	Text/Options
1. Name	Text (full name)
2. Email address	Email
3. Project Name / ESA's funded activity	Text (project name)
4. Brief Project Description	Short text
5. Date for meeting	Booking calendar
6. What resource do you want to publish?	Options: only data, only scientific workflow / code, both

7. Title of your dataset / research outcomes	Text (public title)
8. Short description of your dataset/workflow (meaningful to external users)	Short text (plain language)
Dataset Information	
9. Who will use your data and how?	Short text (user groups)
10. What is your data's main purpose?	Options: visualisation, analysis, both
11. What type of data is it?	Options: trajectories, gridded phenomenon, vectors/features, point observations, time series, model output, other (text)
12. Spatial extent of your dataset/research outcomes (region + bbox)	Text (region + bbox)
13. Temporal extent of the dataset (start : end date)	Date range
14. What access patterns should we optimise for?	Description
15. What libraries/ecosystems do you work with?	Descriptions
16. Are you open to changing the data (format/structure/metadata) to align with best practices?	Options: Yes, No, Maybe, Depends (text)
17. Would you require hosting your data in EarthCODE for long-term preservation?	Options: Yes, No, Not now, but in the future, I do not know
18. What is the size of your datasets (output data + code/workflow)?	Text (e.g., GB/TB)
19. Please specify the format of your dataset (GeoTIFF, Python, NetCDF, Zarr, others)	Text (formats + versions)
20. Does your dataset have restrictions related to accessibility? What is the license?	Text (license + access)
21. Final version of your dataset — link	URL

Experiment / Workflow Information	
22. Code / Workflow — link to public GitHub repository or other location	URL
Additional Information	
23. Documentation — link to papers/handbooks/guides describing outcomes	URL
24. Other comments to EarthCODE team	Long text
25. Consent statement (tick to confirm you agree to processing under ESA Privacy Policy)	Checkbox + URL
26. Do you have a platform preference to implement this work?	Drop down list box

3 D.2. ENGINEERING SUPPORT SERVICE REPORT

This report provides a summary of progress:

[Engineering Support · Engineering and Product Update Services](#)

It is based on the following reporting format:

Engineering Support Service Report – Finance						
Service ID	Service Name	Progress %	Hours Spent	Hours Remaining	Total Budget	RAG
ENG001	Engineering Service 1					
ENG002	Bulk Engineering Service 2 - Eng. Work 1 - Eng. Work 2 - Eng. Work 3 - Eng. Work 4 - Eng. Work 5					

In addition, the following templates can also be used to report progress:

- Engineering Data Report

https://github.com/ESA-EarthCODE/OSC-Entries/blob/main/.github/ISSUE_TEMPLATE/engineering-data-report.md

- Engineering Workflow Report

https://github.com/ESA-EarthCODE/OSC-Entries/blob/main/.github/ISSUE_TEMPLATE/engineering-workflow-report.md

4 D.3. SCIENCE PRODUCTS UPDATE SERVICE DEFINITION

This service concerns the provision of new products by a platform using an existing workflow. It supports the provision of an **operational service** against a defined SLA. This service can configure an existing workflow to provide new products, but workflow software changes should be supplied by the Engineering Support Definition Service.

Requests for this service are likely to come from ESA or the EarthCODE consortium using the following link [Product Updates · Engineering and Product Update Services](#).

This stream of work may also be enabled by solutions that support operational services across platforms. These solutions might measure service availability, service reproducibility, and platform/container compatibility etc.

4.1 Science Products Updates Service Lifecycle

The following Science Products Updates Service Lifecycle is used:

1. Science Products Update request is added to the board ([Product Updates · Engineering and Product Update Services](#))
2. Request is authorized by ESA
 - 2.1. ESA sends WO request by email (may include from the start the details provided in the form and agreed with the subcontractors involved)
 - 2.2. Offer is provided to ESA by TPZ via response to WO, including the cost.
 - 2.3. The WO is signed by ESA TO, CO and TPZ, and exchanged by email.
3. Request is fulfilled and the GitHub project is used to provide progress updates by the service provider
4. Request is completed and a Product is published or pre-published to the live EarthCODE environment

The work order procedure is defined in the appendix to CCN-4, an excerpt of this is provided for information in Appendix A – Work Order Procedure.

The Service Manager role is responsible for managing this overall lifecycle.

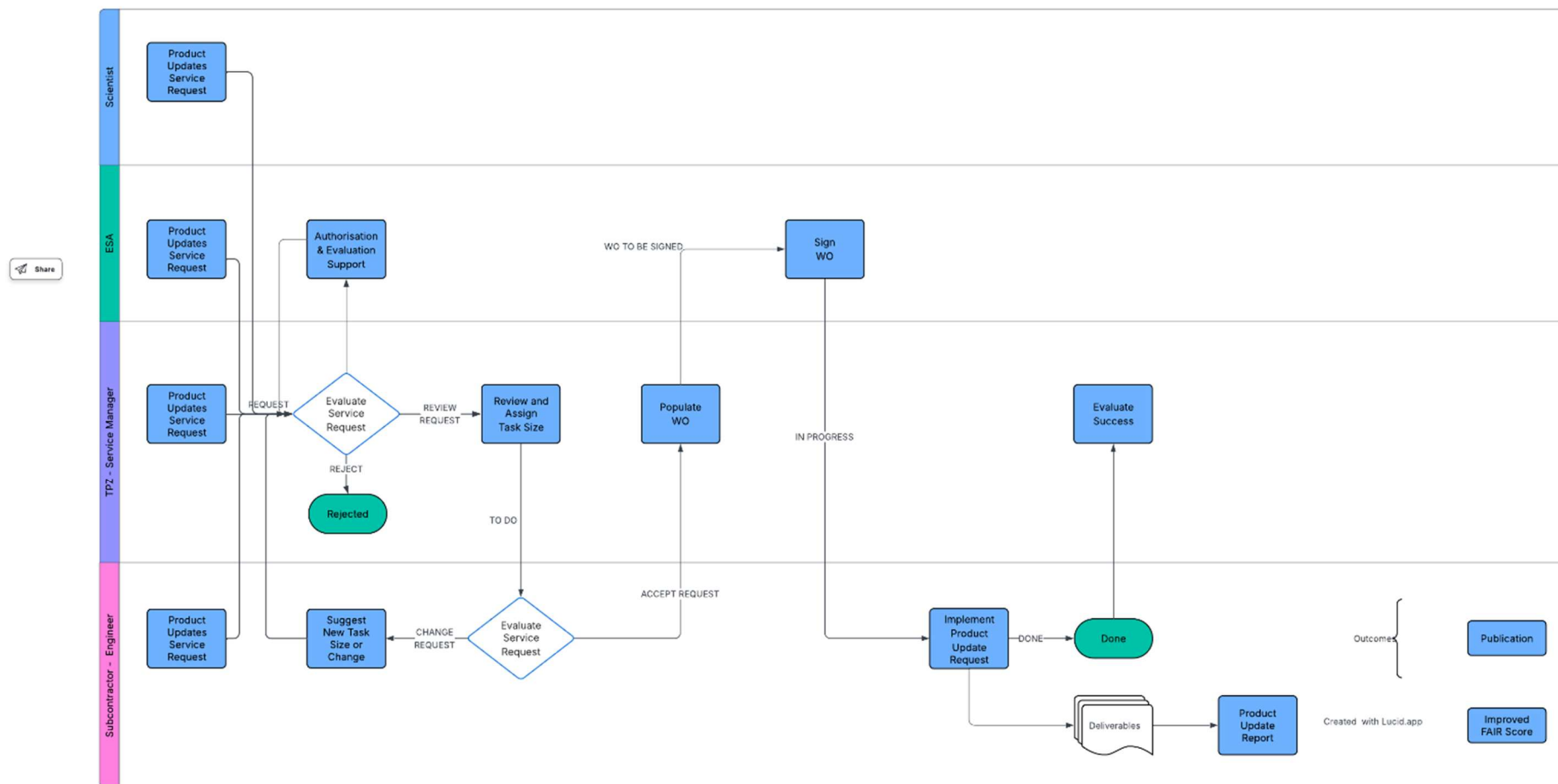


Figure 2 : EarthCODE - Science Products Updates Service Lifecycle

4.2 Product Update Operational Services

Size	Description (P1-P4 definition provided later)	Range for Days of Work (can be converted to hours)
S	Confirm a named platform can fulfil an operational service request and the cost of service provision (P1 and P2)	1 to 3 days
Bespoke	Deployment and provision of initial service (P3)	Defined below
Bespoke	Ongoing Support (P4)	Defined below

4.3 Product Update Operational Services Definition

This service is concerned with providing regular product updates to an existing product using an existing workflow. It concerns the provision of an operational service and not changes to a workflow in order to provide an operational service. Any workflow changes need to be made by the Engineering Support Service.

Science Products Update Service Definition	
Product Update Service Provision	
Task	Description
P1 - Pre Operational Service Technical Analysis	Confirm requirements are clear; the workflow is ready to be used subject to configuration and the subcontractor's operational environment can meet requirements.
P2 - Pre Operational Service Cost Analysis	Confirm the initial total cost to deploy the service. Breakdown to include i) Cost of deployment ii) Additional Set-up costs Confirm the total cost per month of service provision for the duration of the service. iii) Cost of storage iv) Cost of processing v) Cost of support vi) Any other costs.
P3 - Provision of Initial Service	Configure and deploy service and provide a release note and update Operations Manual. Provide SLA.
P4 - Ongoing Support	Provide Service Operations Report and monitor SLA to ensure service performs correctly. Inform TPZ of any SLA exceptions.

4.4 Customer Products Update Service Request Form

The following template is used to request Product Update Service requests:

https://github.com/ESA-EarthCODE/OSC-Entries/blob/main/.github/ISSUE_TEMPLATE/product-update-request-template.md

It was based on the form below and can be updated further as the service evolves.

Product Update Request ID	PROD0001
Product Definition Summary	(text / options)
1. Name	Text (full name)
2. Email address	Email
3. Project Name / ESA's funded activity	Text (project name)
4. Brief Project Description	Short text
5. Date for meeting	Booking calendar
6. What published product do you want to be updated?	
7. What is the nature of the update and the frequency of the update?	
8. What is the size of your new products during this period?	Text (e.g., GB/TB)
9. What is the termination date for this service to stop creating new products?	
10. Do you have a platform preference to implement this work?	Drop down list box

5 D.4. SERVICE OPERATIONS REPORT

This report provides a summary of progress:

[Product Updates · Engineering and Product Update Services](#)

It is based on the following reporting format:

Service Operations Report – Finance						
Service ID	Service Name	Progress %	Hours Spent	Hours Remaining	Total Budget	RAG
PROD001	Prod Update Service 1					
PROD002	- Prod Update Service 1					

In addition, the following template can also be used to report progress:

- Product Update Report

https://github.com/ESA-EarthCODE/OSC-Entries/blob/main/.github/ISSUE_TEMPLATE/product-update-report.md

6 APPENDIX A – WORK ORDER PROCEDURE

WORK ORDER PROCEDURE

All Agency orders for Additional Activities shall be placed by means of Work Orders (WO), a sample of which is attached in Annex hereto, and in accordance with the procedure described below. It is recalled that Changes to this Contract, including in particular the addition or removal of any subcontractor, require a Contract Change Notice signed by the Agency's Contracts Officer and Technical Officer. Changes made in any other form, be it orally, via Work Order or other, are invalid.

1.If applicable after asking for quotation and receipt of an accepted offer, or else without Request for Quotation and offer, the Agency shall raise a WO and complete all sections of the WO form on the basis of the information contained in the Contract, its Appendices and CCNs and the offer, answering to the Agency's Request for Proposal (RFP).

If the WO involves one or more contractually agreed Sub-Contractor(s), this should be addressed in the "Price" and "Payments" sections of the WO form.

The Work Order shall mention the Work Order Firm Fixed Price calculated on the base of the FUPs (Hourly Rates) agreed below, amount which is to be taken from the Ceiling Price stated in CCN4. The Work Order shall further state the remaining available Ceiling Price after the Work Order.

2. The completed WO form shall be signed by the Agency's Technical Officer of the Contract and sent to the Contractor for countersignature.

3. In the event of any conflict between the WO itself and its annexes and/or documents referred to therein, the WO shall take precedence.

4. On receipt of the WO, the Contractor's Technical Representative, as mentioned in Clause 5.2a) of Article 5 of the Contract, shall immediately sign the WO, return a signed copy to the Agency's responsible Technical Officer, keeping in copy for information only the Agency's responsible Contracts Officer, and implement the WO.